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Do auditor resignations reduce uncertainty about the quality of firms' financial reporting?

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Abstract

We assess the conditions under which auditor resignations reduce uncertainty about the quality of financial reporting of former and continuing clients of the resigning auditor. Our analysis is based on a sample of 109 auditor resignations in the period 1994–1998. Approximately one-quarter of these resignation announcements are accompanied by disclosure of a disagreement over accounting treatment or over the adequacy of internal controls—factors that point toward diminished credibility of former client's financial statements—and are associated with negative abnormal returns for the former client. In contrast, we find no stock market effect for the remaining firms (i.e., three-quarters of our sample) for which the auditor resignations are "unexplained", suggesting that the act of resignation, by itself, is not informative. Further, we find that these unexplained resignations, which likely result from auditors' periodic risk-related reviews of their client portfolios, transfer value-relevant information to another class of investors. In particular, we find that industry matched *continuing* clients of the auditor that

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also have poor performance experience *positive* abnormal returns in the days surrounding media-announced resignations from former clients. This suggests that in an environment where poorly performing, high-risk audit clients are screened contemporaneously, an auditor's unexplained resignation from one client, but not from a similar client in its portfolio, helps reduce uncertainty about the credibility of *continuing* clients' financial reporting.
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1. Introduction

This paper investigates the conditions under which auditor resignations provide information to investors, and the nature of the information conveyed by resignation announcements. Our investigation extends prior research in two ways. First, prior research characterizes resignation announcements as bad news for investors, but does not specify what information (if any) is conveyed by the act of resignation. We predict and find that auditor resignations are only informative when the announcements of resignations contain specific information indicating diminished reliability for the former clients' financial statements. Similarly, we predict and find that the act of resignation by itself is not informative because auditors resign from firms that experience deteriorating financial performance, and skeptical investors are likely to have already discounted those firms' prospects. We show that our inferences differ from those in the extant literature in part because prior research has failed to systematically control for other information that is contemporaneously released. Second, we make predictions about the informational value of resignation announcements for continuing clients of the resigning auditor, a class of firms not analyzed in prior resignation research. We predict and find that, when resignations likely result from auditors' centralized risk review of their portfolio of clients, the resignation announcement triggers a positive information transfer for continuing clients that are experiencing deteriorating financial performance and that are in the same industry as the former clients.

We assess whether auditor resignations reduce uncertainty about the quality of firms' financial reporting using a sample of 109 auditor resignations from publicly traded companies over the period 1994–1998.[1] As in prior research, we find that auditors typically resign from firms that are experiencing poor recent financial performance (DeFond et al., 1997; Krishnan and Krishnan,

[1] We use "the quality of financial reports" to designate the degree to which investors are willing to rely on reported financial data in making investment decisions. We use quality and credibility of financial reports interchangeably.

1997). This feature of firms from which auditors resign is important to our predictions because prior research has also shown that investors are skeptical of the financial disclosures of poorly performing firms.[2] In particular, our conjecture that investors require a premium for holding the stock of poorly performing firms to compensate them for greater uncertainty about the quality of such firms' financial reporting has two implications. First, for resignation announcements to negatively affect stock prices, the information accompanying the resignation must revise investors' beliefs about the quality of a firms' financial reporting in excess of the credibility discount embedded in price. Second, if investors pool continuing clients of the auditor who, like former clients, experience poor stock price performance prior to the resignation as having poor quality financial reporting, the *absence* of resignation from those clients has the potential to revise upwards investors' perception of the credibility of continuing clients' financial reports.

Our sample includes two types of resignations. *Attributed* resignations represent approximately one quarter of our sample and include an accompanying disclosure that attributes the resignation to a disagreement over accounting treatment or over the adequacy of internal controls.[3] We predict that in cases of *attributed* resignations, the reason for the resignation points toward diminished reliability for the former client's financial statements (SEC, 1988). We find that these *attributed* resignations are associated with negative abnormal returns. This result is consistent with prior research that has shown that such reportable events have information content (DeFond et al., 1997; Whisenant et al., 2003).

The majority of our sample resignations are *unexplained*. In the 1990s audit firms initiated routine risk reviews of their portfolios of audit clients, and resignations occurred because auditors determined that specific audit clients sustained increases in audit risk and/or in business risk for which the auditors were not being compensated (MacDonald, 1997; Jones and Raghunandan, 1998; Winograd et al., 2000; Bell et al., 2002; Johnstone and Bedard,

[2] Investor skepticism is consistent with research findings indicating that declining performance (1) increases the likelihood that managers make relatively aggressive accounting and reporting choices (e.g., Petroni, 1992; Sweeney, 1994; Beneish, 1997), and (2) is often associated with illegal corporate behavior (National Commission on Fraudulent Financial Reporting (NCFFR), 1987; National Association of Certified Fraud Examiners (NACFE), 1993.

[3] The Securities and Exchange Commission (SEC, 1988, 1989) requires registrants to report (in a Form 8-K) auditor changes within five business days of the date of registrant's disassociation with the auditor and to indicate whether the disassociation was client-initiated (i.e., a dismissal) or auditor-initiated (i.e., a resignation). Disclosure of the auditor's reason for resigning is not required, but SEC Financial Reporting Release (FRR) No. 31 does require disclosure of information about accounting disagreements, internal controls deficiencies and financial statement reliability "reportable events" when accompanied by auditor changes (SEC, 1988).

2004).[4] Because we exclude resignations that occur for innocuous reasons (mergers, retirement of an auditor, relocation of headquarters), we propose that these unexplained resignations most likely result from auditors' periodic risk reviews of their portfolios of audit clients. We predict and find that in cases of *unexplained* resignations the standard boilerplate language in the Form 8-K filing does not convey new information to investors, who are likely to have scrutinized these former clients and have already discounted their future prospects using alternative information sources.

An unstated but apparently maintained presumption in prior auditor resignation studies is that the act of resignation conveys information not known and not anticipated by investors. Our results question this presumption: resignations announcements only reduce uncertainty about the quality of former clients financial reporting when the announcement *specifically provides information on the nature* of the accounting disagreement. *Unexplained* resignations are not informative because investors are skeptical of former clients' financial reports, having observed, prior to the resignation announcement, events that indicate financial difficulty (losses, dividend cuts/omissions, asset sales, labor force reductions, refinancing, debt covenant default, and lawsuits).

We propose that the auditor risk-review process could provide value-relevant information for continuing clients. In particular, in an environment where poorly performing, high-risk audit clients are screened contemporaneously, the *absence* of a resignation may convey private information about continuing clients that are in the same industry as the former client and that have had similar recent stock market performance as the former client. Therefore, we predict that unexplained auditor resignations can assist investors in separating worse quality firms (i.e., former clients) from better-quality firms (i.e., continuing clients) and that continuing clients should experience positive abnormal returns in the days surrounding the public announcement of auditors' unexplained resignations from former clients.

Our tests are based on a sample of continuing clients in the same industry that have the most similar (i.e., poor) stock price performance in the 15-month period preceding the resignation announcement. Consistent with the notion of a pooling equilibrium, we find that continuing clients appear healthier than former clients (they are larger, more profitable, less levered and have lower risk of bankruptcy), yet their prior stock market performance is similar to that of former clients, suggesting that investors discount the credibility of continuing clients' financial reports.

[4] In an attempt to reduce potentially costly exposure to a subset of the riskier clients, all large audit firms implemented these centralized, annual business-risk-based reviews of their client portfolios during the 1990s. For example, PricewaterhouseCoopers' risk management process "has helped the firm…terminate relationships with certain clients when sufficient mitigation of identified risks seems unlikely" (Winograd et al., 2000, p. 177).

Our stock market analysis reveals positive abnormal returns for continuing clients, but only when the unexplained resignation is disclosed in the news media. The evidence is consistent with a scenario where the auditor risk-review process provides information about the credibility of continuing clients' financial statement disclosures so long as the information about the resignation is widely disseminated.

A competing explanation for a positive stock-price reaction for continuing clients, based on mean effects documented in prior intra-industry information-transfer research, is as follows:[5] a positive stock-price reaction could be observed because auditor resignations signal declining prospects for former clients and enhanced opportunities for surviving companies in the same industry. To assess this alternative explanation, we evaluate the stock price reaction to auditor resignation announcements for a sample of firms that are matched to former clients on industry and stock-price performance, while ensuring that the matched firm has a different auditor from the former client (i.e., we obtained this sample using the same criteria we used to identify continuing clients, except that in this case we require the matched firm to be audited by somebody other than the resigning auditor). We find that the stock price reaction for this matched sample is not significantly different from zero. We conclude that the continuing clients' positive returns cannot be explained as the result of reduced competition in the continuing clients' industries and that the auditor resignation conveys positive information about the financial reporting credibility of the continuing clients.

Empirical leverage for our analysis comes from two sources. In addition to controlling for potentially contaminating information that is concurrently released in either Forms 8-K or in the news media, we show that our findings are robust to using alternative return specifications, and that they are not an artifact of firms with low stock prices for which a change in spread leads to large returns. Our findings suggest that auditor resignations triggered by accounting system disputes provide information about former clients and that the auditor risk-review process provides information about continuing clients. In both cases, the information conveyed by resignations relates to the credibility of the firm's financial reporting.

The remainder of the paper is organized in four sections. Section 2 motivates and develops our predictions. Section 3 describes our research design. Our empirical results are discussed in Sections 4 and 5 concludes the paper.

[5] This research studies stock-price reactions for multiple firms within an industry and interprets the price reactions to be indicative of the overall health of the industry or of shifts in the competitive balance within the industry. Prior research has investigated numerous such settings including announcements of earnings (e.g., Foster, 1981), management forecasts of earnings (Baginski, 1987), sales (e.g., Olsen et al., 1985), industrial accidents (e.g., Bowen et al., 1983), mergers (Eckbo, 1983), and dividend changes (Firth, 1996).

2. Background and hypotheses development

This section discusses prior research on the stock price effects of resignation disclosures, and develops hypotheses for the effect of resignation announcements for former clients, for same-industry continuing clients of the resigning auditor, and for same-industry continuing clients of a different auditor.

2.1. Prior research on stock price reactions to auditor resignations

Prior research on auditor resignations has generally taken one of two approaches: examination of auditor and former client characteristics that are associated with resignations (e.g., Krishnan and Krishnan, 1997; Shu, 2000) and examination of the stock price reaction at the time of resignations (e.g., Wells and Loudder, 1997; DeFond et al., 1997; Shu, 2000; Whisenant et al., 2003).[6] We take the latter approach and attempt to identify the conditions under which auditor resignations lead to stock market reactions for audit clients.

Wells and Loudder (1997) report a significantly negative reaction (−0.66% average prediction error) for a sample of 82 firms during 1988–1991 in the two-day event period beginning on the date the Form 8-K containing initial information about resignations is filed. In additional analyses, they include dummy variables for the contemporaneous disclosures of the reasons for the resignation or other information related to the resignation (e.g., naming a successor auditor). They find that the coefficients on these dummy variables are insignificant, and conclude that it is the "disclosure itself, not any other related information, that is causing the results" (p. 143). DeFond et al. (1997) examine 37 resignations occurring between 1982 and 1987, and find significantly negative returns for the period between the auditors' resignations and the receipt of the related Form 8-K by the SEC and for the five-day period starting on the SEC receipt date (median excess returns of −5.3% and −1.7%, respectively). They also consider the presence of potentially confounding news in the Form 8-K filings, and conclude that confounding news events did not affect returns around the time of the resignation announcement. Shu (2000) considers how litigation risk and clientele-adjustment proxies are associated with auditor res-

[6] Many earlier studies investigated auditor *changes* (e.g., DeFond, 1992; Francis and Wilson, 1988; Fried and Schiff, 1981; Johnson and Lys, 1990; Klock, 1994; Nichols and Smith, 1983; Schwartz and Soo, 1996; Smith, 1988), but relatively few have examined auditor resignations, per se. This lack of research largely is the result of the difficulty in identifying auditor resignations before the issuance of FRR No. 31 (SEC, 1988). Prior to 1988, companies were required to file a Form 8-K to disclose a change of auditor, but were not required to disclose whether the change was the result of a resignation or a dismissal. After 1988, investors can distinguish resignations from dismissals because the SEC requires explicit 8-K disclosure of the exact types of auditor changes experienced by registrants.

ignations, and incorporates a market reaction analysis on a subset of her sample. In her market-reaction sub-sample, she reports a −3.1% excess return for the three-day window centered on the Form 8-K filing date. She reports that "only a couple cases" might be confounded by other Form 8-K disclosures (Shu, 2000, footnote 24, p. 195).

Whisenant et al. (2003) find significantly negative returns associated with Form 8-K disclosures of FRR No. 31 reportable events filed by 118 companies from 1993 to 1996. They report significantly negative CARs for the subset of firms that had auditor resignations *and* reportable events ($n = 57$, with an average three-day CAR of −4.48%) but do not provide descriptive returns results for all resignation observations. Their regression results indicate that the presence of reportable events and auditor–client disagreements are associated with negative abnormal returns even beyond the effect of the resignation disclosure itself. The resignation disclosure is significantly associated with negative abnormal returns when using seven-day window returns (i.e., $[-5, +1]$), but not when using three-day window returns (i.e., $[-1, +1]$).

Surprisingly, prior research has either not controlled for a comprehensive set of sources of potentially contaminating concurrent information (e.g., Shu, 2000; Whisenant et al., 2003 controlled for information reported in the 8-K, but not for information reported in the news-media) or has found a very low occurrence rate for concurrent news.[7] We find that auditor resignation announcements are accompanied by media-released information about earnings, management turnover, losses, dividend cuts, and adverse legal proceedings for 26 firms (21% of our sample). It appears particularly important to consider this potentially contaminating information in our sample; excluding the contaminated observations reduces the overall stock price reaction to auditor resignations by 34%. We therefore restrict our analyses to companies for which no other news reports were found during the days around the resignation disclosure.

Prior research has examined several additional characteristics that might be associated with stock price reactions to auditor resignations. These characteristics can be classified into two groups: characteristics of the resignation disclosure and characteristics of the former client. Within the first group, we consider the source of the initial public announcement of the resignation (i.e., Form 8-K versus media disclosure) and the timing of the resignation relative to the fiscal

[7] Wells and Loudder (1997, footnote 7) find two instances among their 86 firms with concurrent information released at the same time as the resignation announcements. DeFond et al. (1997, p. 39–40) find two of their 37 resignation firms had contemporaneous news other than that now required in Form 8-K filings for their "pre-filing period". Shu (2000, footnote 24) found "only a couple cases" (out of 135 resignations) where auditor resignation filings may be confounded by other disclosures in the 8-K.

year-end of the former client. Within the second group of characteristics we consider the presence of a prior going-concern opinion by the auditor and the likely mismatch between the former client and auditor (cf., Shu, 2000). We improve on prior research by explicitly controlling for the effect of FRR No. 31 reportable events accompanying the SEC-mandated resignation disclosures when examining the association between the stock price reaction to resignation announcements and these characteristics. This is important because resignations that are accompanied by the disclosure of reportable events are likely to be informative to investors about former clients (Whisenant et al., 2003), whereas resignations that are not accompanied by such disclosures are less likely to provide investors with information that reduces their ex ante uncertainty about the former client's financial reporting credibility.

2.2. Development of hypotheses

Our hypotheses for the effect of resignation announcements relate to former clients and two types of continuing clients: same-industry continuing clients of the resigning auditor, and for same-industry continuing clients of a different auditor. We summarize our predictions in Fig. 1, and discuss their development below.

2.2.1. Former clients

When a resignation is "attributed" to a reportable event, the combined disclosure is likely to be informative to investors. In particular, information about the resignation and its associated cause is likely to have been unknown prior to the public report and this package of information should be important to investors as they evaluate the future cash flows of the former client. In contrast, for "unexplained" resignation announcements (i.e., those *not* accompanied by additional Form 8-K disclosures about the reason for the resignation), the news of the resignation itself is not likely to identify or resolve much uncertainty for investors. Resignations resulting from periodic risk reviews by auditors are likely in this category since no specific events or conditions led to these resignations. General increases in risks to the auditors are probably anticipated by investors based on alternative information sources, so without auditors' specific identification of causes for the resignations, little or no uncertainty is resolved about former clients' financial reporting credibility. All else equal then, we do not expect significant former-client stock price changes for unexplained resignation announcements.

H1a. *Attributed* auditor-resignation announcements (i.e., those accompanied by information about reportable events or accounting disagreements) are associated with negative abnormal returns for former clients at the time of the resignation announcement.

Reason for Resignation	*Former client*	*Continuing clients of the resigning auditor matched on industry and performance*	*Continuing clients of a different auditor matched on industry and performance*
Attributed ***(unreliable internal control, accounting disagreement)***	***Stock Price Reaction (SPR)<0*** **[H1a]**	***SPR=0*** **[H2a]**	***SPR=0*** **[H3]**
Unexplained	***SPR=0*** **[H1b]**	***SPR>0*** **[H2b]**	***SPR=0*** **[H3]**

Fig. 1. Predictions: In the first two hypotheses, our predictions depend on the stated reason for resignation. For former clients, we predict that (1) for *attributed resignations*, disclosures of disagreements over accounting treatment or disclosures of unreliable internal controls point to diminished reliability in financial statements and are associated with a negative stock price reaction, and that (2) *unexplained resignations*, likely the result of auditors' centralized risk-review process, convey no new or unanticipated information to investors. For continuing clients of the resigning auditor, (1) we predict no information transfer in the case of *attributed resignations* because the resignation identifies an issue that is client-specific, and (2) to the extent that *unexplained resignations* result from auditors' contemporaneous risk reviews of former and similarly performing continuing clients in the same industry, we predict a positive information transfer for the continuing clients. Our third hypothesis (stated in null form) suggests that the resignation announcement does not affect the level of competition in the former client's industry. We test this hypothesis by examining whether there is an information transfer to continuing clients of a *different* auditor matched on industry and performance.

H1b. *Unexplained* auditor-resignation announcements (i.e., those *not* accompanied by information about reportable events or accounting disagreements) are not associated with abnormal returns for former clients at the time of the resignation announcement.

2.2.2. Continuing clients

Our second set of hypotheses addresses "continuing clients"—companies audited by the same auditor, operating in the same industry as the former client, and with similar prior stock price performance to the former client, but from which the auditor did *not* resign. A resignation from a former client, coupled with the absence of a resignation from a similar continuing client, has potential to provide news to investors about the continuing client. However, the extent to which news is contained in the resignation depends on whether the resignation from the former client was triggered by an idiosyncratic, client-specific event or condition (i.e., an *attributed* resignation) or whether the resignation is most likely the result of a centralized risk-based review conducted by the auditor (i.e., an *unexplained* resignation).

Because there is no way for investors to disentangle the information contained in the reportable event from the information contained in the attributed resignation, investors cannot infer anything about a possible internally conducted risk-based screen performed by the auditor. Therefore, we propose that an attributed resignation from a former client does not result in abnormal returns for the continuing client.

In contrast, we propose that an *unexplained* resignation from a former client is likely the result of a centralized risk-based evaluation conducted by the auditor. By initiating an unexplained resignation from the *former* client, the resigning auditor is revealing the type of company targeted or identified by its risk-based evaluation. Because the auditor chooses to maintain its association with the *continuing* client (that investors assessed *ex ante* to be similar in risk characteristics to the former client), investors can infer that the continuing client is a better "type" than the former client. Therefore, investors in continuing clients can interpret an unexplained resignation as providing evidence that the risks for the continuing client are lower than previously assessed, leading to a positive price response.

H2a. *Attributed* announcements of auditor resignations from former clients are not associated with abnormal returns for continuing clients of the auditing firm that resigned from the former client.

H2b. *Unexplained* announcements of auditor resignations from former clients are associated with *positive* abnormal returns for continuing clients of the auditing firm that resigned from the former client.

To the extent that the resignation announcement from the former client might signal poorer prospects for the former client, a positive price reaction for continuing clients might result from improved future prospects caused by reduced competition in the former and continuing clients' industry. We explore this alternative explanation by considering an additional sample of firms matched to former clients on industry and prior stock price performance (i.e., the same criteria used to match continuing clients to former clients), *but audited by a different auditor* (hereafter "industry-matches"). We do not expect our continuing client prediction—that the resignation by the auditor from the former client should be viewed as positive news for investors in similar continuing clients of the same audit firm—to extend to economically similar clients audited by *other* firms.

H3. Announcements of auditor resignations from former clients are not associated with abnormal returns at the time of the resignation announcement for similar continuing clients audited by other audit firms.

3. Research design

3.1. Sample selection

We identify auditor resignations using the *Auditor Trak* database (the database). In January 1994, the database (excerpts of which are published in the Public Accounting Report) began keeping track of auditor resignations. Through February 1998, the database provides the following information for 689 instances of auditor resignations: (1) the name of the firm whose auditor resigned, (2) the name of the resigning auditor, (3) the name of the newly hired auditor (when available), (4) the type of opinion in the most recent audit, and (5) the date of the auditor's letter of resignation.[8]

Because we require that data be available to estimate the stock market reaction to the announcement of the auditor resignation and to identify the reason for the auditor resignation we reduce the initial sample of 689 firms down to 109 firms. As summarized in Panel A of Table 1, we eliminate:

1. Three hundred and twenty firms that do not appear on any of the CRSP NYSE, ASE and OTC files.
2. One hundred and twenty four firms that were delisted prior to the date of the auditor resignation.[9]
3. Eleven firms from which the auditor resigned after the last date included in our CRSP file and 6 firms that started trading after the date of the auditor resignation.
4. Sixty eight firms for which we cannot identify an event date from either the news media (search described below) or from SEC filings. Our search for Form 8-K filings included (1) the Lexis/Nexis 8-K sub-library of the COMPNY library, the SEC Edgar Archives, and the Indiana University Q-File collection. In addition, our Q-File supplier was unable to identify Form 8-K filings for these 68 firms in a 6-month period around the *Auditor Trak* database's date for the auditor resignation.

[8] According to Mr. Brian McGreevy of the *Public Accounting Report*, the information contained in the *Auditor Trak* database was drawn from Form 8-K filings with the SEC. The SEC improved the timeliness of Form 8-K disclosure in March 1989 when it issued *FRR No. 34*, requiring companies to file an 8-K within *five* days of being notified of a resignation by the auditor. In March 2004, the SEC again improved the timeliness of resignation disclosures, requiring companies to file an 8-K within *four* days of being notified of a resignation that occurs on or after August 23, 2004 (SEC, 2004).

[9] Analysis of the relevant CRSP codes reveals that delisting occurred because the registrants had insufficient capital or failed to meet financial guidelines (82 firms), were delinquent in filing (20 firms), merged (7 firms), or because of some other reason (17 firms). The most common reason for delisting is consistent with the results of extant research suggesting that auditors' resignations are associated with deterioration in clients' financial condition (Krishnan and Krishnan, 1997; Schwartz and Soo, 1995).

Table 1
Sample selection and attributes

Panel A: Sample selection		
Resignations identified in data base	689	
Minus		
Firms that are not on CRSP	320	
Firms that delisted prior to resignation	124	
Firms that started trading after resignation	6	
Unavailable returns on CRSP	11	
Resignations for which the event date is not available	68	
Firms with insufficient returns data in estimation period[a]	37	
Reason for resignation is in the "Other" category[b]	14	
Final sample	109	
Panel B: Reason for resignation[b]		
	Frequency	Percent
No reason disclosed	79	72.5
Specific accounting and auditing concerns	30	27.5
Total	109	100.0
Panel C: Industry classification based on 2-digit SIC codes		
10–19 Extraction and construction	5	4.6
20–29 Commodity production	14	12.8
30–39 Manufacturing	21	19.3
40–49 Utilities and transportation	13	11.9
50–59 Wholesale and retail	18	16.5
60–69 Financial services	11	10.1
70–79 Business services	20	18.3
80–89 Health and other services	7	6.4
Total	109	100.0
Panel D: Resigning auditor		
Arthur Andersen	10	9.1
Coopers and Lybrand	26	23.9
Deloitte and Touche	9	8.3
Ernst and Young	13	11.9
KPMG Peat Marwick	11	10.1
Price Waterhouse	12	11.0
BDO Seidman	4	3.7
Grant Thornton	2	1.8
McGladrey and Pullen	4	3.7
Other	18	16.5
Total	109	100.0

[a] These are firms with less than 100 observations during the estimation period of 200 days (either $(-500, -301)$ or $(+61, +261)$), over which we estimated market model parameters.

[b] The reason for resignation is identified in the media or 8-K report which first announces the resignation. The "specific accounting and auditing concerns" reason applies to all firms that cited accounting-related disagreements with the auditors or where the auditors advised the company (i) that internal controls necessary to develop reliable financial statements did not exist, or (ii) that they were unwilling to rely on management's representations. The "Other" reasons include resignations because of auditor retirements, sales of office, or independence issues.

5. Thirty seven firms with missing returns in the estimation period.
6. Fourteen firms that resigned but the resignations were due to non-client factors (e.g., auditor retirements, sales of practice offices, or independence issues).[10]

For our remaining 109 sample firms, we used Forms 8-K and media reports related to the auditor resignations to determine the reason for their resignation and the earliest date that the auditor resignation became public information. We conducted our searches by examining 8-K filings and media announcements in a 6-month period around the *Auditor Trak* database's date for the auditor resignation. For a majority of the remaining firms (79 firms, or 73%), the 8-Ks or media reports indicate that the auditor resigned but offer no reason for the resignation. These 8-Ks generally use boilerplate language indicating that there are no disagreements on any matter of accounting principles or practices, financial statement disclosure, or auditing scope or procedure.[11] For the remaining 30 firms (28%), we are able to identify reasons that indicate there were accounting-related disagreements with the auditors and/or that the auditor advised the company that internal controls necessary to develop reliable financial statements did not exist or that the auditor was unwilling to rely on management representations.[12]

[10] For example, Rocky Mountain Internet, Inc., in an 8-K filed January 28, 1997, reports that: "Effectively, the Registrant's former independent accountant, McGladrey and Pullen, LLP, simultaneously resigned as of January 20, 1997. The Denver office of McGladrey and Pullen was acquired by Baird, Kurtz and Dobson on June 17, 1996. All former audit engagement members are now with Baird, Kurtz and Dobson, and will continue to be involved with the Registrant's Fiscal 1996 audit".

[11] For example, First Cash, Inc., disclosed the following in its 8-K report about the auditor's resignation: "On May 28, 1997, Price Waterhouse LLP resigned as the independent accountants for the audit of the Company's financial statements and the client-auditor relationship between the Company and Price Waterhouse LLP ceased. The reports of Price Waterhouse LLP on the financial statements of the Company for each of the past two years contained no adverse opinion or disclaimer of opinion or any qualification or modification as to uncertainty, audit scope or accounting principles". The 8-K report includes a letter from Price Waterhouse LLP indicating that they agree with the statements contained in the 8-K.

[12] For example, Fiberchem, Inc, in an 8-K filed on January 31, 1997, reports that: "On January 24, 1997, KPMG Peat Marwick LLP (the 'Former Accountant') resigned as FiberChem, Inc.'s (the 'Company') principal accountants. The Former Accountant [Peat Marwick LLP] did not state any reason for resigning in its resignation letter to the Company. However, in its letter to the Audit Committee and its Material Weakness letter both dated January 10, 1997 and delivered January 23, 1997, the Former Accountant reported 'Disagreements with Management' on financial accounting and reporting matters and auditing scope concerning revenue recognition that, if not satisfactorily resolved (which all were) would have caused a modification of their report on the 1996 consolidated financial statements. The disagreements aggregating approximately $1,800,000, concerning certain transactions termed "consignments" by the Former Accountant, products warehoused for customers, and a research and development effort, none of which met the requirements for revenue recognition under generally accepted accounting principles".

Panel C of Table 1 shows the industry groupings of our sample firms. About 19% of the firms (21 firms) are manufacturers (codes 30–39), 18% (20 firms) are software and service firms (codes 70–79), and 17% (18 firms) are wholesale and retail firms (codes 50–59). In total, 39 different 2-digit SIC industries and 76 different 2-digit SIC industries/years are represented in our sample, suggesting that industry/time clustering is not likely to be an issue in our sample. Panel D of Table 1 shows the identity of the resigning auditor. About 74% of resigning auditors are Big Five firms (or their predecessors) and Coopers and Lybrand accounts for 24% (26 firms) of sample resignations.

Sample firms range in size (i.e., market value of equity) from Purus, Inc. ($1.35 million) to IDB Communications Group ($1252 million). In Table 2, Panel A, we contrast the 109 former clients with the matched continuing clients (i.e., firms matched by auditor, industry, and recent stock price performance). We are able to match 83 out of 109 firms using these criteria.[13] The data presented in Table 2 are measured in the fiscal year prior to the fiscal year in which the auditor resignation occurs. We find that the performance of continuing clients, while poor (median average daily compounded abnormal return = −0.00091), is better than that of former clients (median average daily compounded abnormal return = −0.00289). Consistent with the notion of a pooling equilibrium, we find that continuing clients appear healthier than former clients (they are larger, more profitable, less levered and have lower risk of bankruptcy), yet their stock market performance is also poor, suggesting that investors discount the credibility of continuing clients' financial statement disclosures.[14]

3.2. Method

3.2.1. Identification of announcement date and contaminated announcements

To determine the earliest date that market participants became aware of the auditor resignation, we performed an extensive news media search. Specifically, we searched all of Dow Jones News Retrieval Service and the Lexis/Nexis COMPNY/ALLNWS library using the following search criteria for each of the 109 companies in our sample "((audit or auditor or auditors) near10 (re-

[13] The continuing client sample is reduced to 83 observations because (1) in 17 instances the auditor for the former client is coded in the "other auditor" category on Compustat, which prevents us from finding a match on auditor, and (2) in 9 instances we find no firm in the industry (same 2-digit SIC) audited by the same auditor and with all necessary stock price information.

[14] Our matching on prior stock market performance produces a sample of continuing clients that is the most similar to the sample of former clients, but does not identify a group of firms with stock market performance that is indistinguishable from former clients. Because we expect information transfers to be more pronounced the worse the continuing clients' performance, our tests are biased against finding positive abnormal returns for continuing clients.

Table 2
A comparison of selected financial characteristics from the year prior to resignation between former clients and continuing clients

Characteristic[a]	Former clients $N = 109$[b] Mean	Former clients $N = 109$[b] Median	Continuing clients $N = 83$[c] Mean	Continuing clients $N = 83$[c] Median	*p*-Value Chi-square test on medians
Panel A: Selected financial characteristics					
Size					
Total assets ($ million)	76.26	27.56	863.11	89.64	0.00
Sales ($ million)	69.75	22.79	724.03	92.92	0.00
Market value of equity ($ million)	78.46	22.92	637.25	126.78	0.00
Leverage and liquidity					
Total debt to total assets (%)	54.70	58.53	47.70	44.93	0.09
Current ratio	2.15	1.46	4.17	2.18	0.02
Profitability					
Return on assets (%)	−9.26	0.64	−2.4	5.87	0.01
Growth					
Sales—growth from last year (%)	15.02	22.47	11.94	12.31	0.06
Other					
Number of shareholders	1599	930	7595	7.59	0.94
Z-score	−2.02	−2.09	−2.76	−2.92	0.01
Life of firm (in months listed)	80	47	120	69	0.04
Returns					
Average daily compounded abnormal returns (−300, −3)	−0.00199	−0.00289	−0.00056	−0.00091	0.01
Panel B: Earnings performance prior to resignation					
Percentage of firms with losses					
In year of resignation (year 0)	56.0%		32.5%		0.00
In years 0 and −1	38.5%		26.5%		0.00
In years 0, −1, and −2	25.7%		15.7%		0.00

[a] Variable definitions:
Total assets = Compustat item #6.
Sales = Compustat item #12.
Market value of equity = Closing price (#24) ∗ Common shares outstanding (#25).
Total debt to total assets = (Total liabilities (#181)/total assets (#6)) ∗ 100%.
Current ratio = (Current assets (#4)/current liabilities (#5)) ∗ 100%.
Return on assets = (Net income (#172) + interest expense (#15))/average total assets ((#6 + lagged #6)/2).
Sales growth = (Sales (#12)/lagged sales (lagged #12)) – 1) ∗ 100%.
Number of shareholders = Compustat item #100.
Z-score is measured as in Zmijewski (1984): −4.803–3.599 ∗ return on assets + 5.406 ∗ debt to total assets −0.1 ∗ current ratio.
Life of firm is measured in months listed (month index of resignation announcement minus month index of firm's first beginning date (begdat) on CRSP).
Market-adjusted returns are computed by subtracting the return on the equally-weighted market portfolio from raw returns.
Earnings = Net income (#172).
Percentage of firms with losses is computed based on net income (#172).

[b] Former clients are firms from which the auditor resigned. The sample size for the former clients ranges from 75 to 109, depending on data availability.

[c] Continuing clients are firms matched to the former clients based on auditor/industry/year and stock price performance over the year prior to resignation, but from which the auditor did not resign. The sample size ranges from 76 to 83, depending on data availability.

sign! or fire! or dismiss!)) and date from 1/1/94 to 3/1/98".[15] Using these data, we determine the day the auditor resignation was first disclosed (Day 0). The search of news sources also identified auditor resignation announcements that were "contaminated" by the simultaneous announcement of potential value relevant news items about the firm in the period $(-1, 1)$. Our contamination analysis reveals that auditor resignations are frequently disclosed alone (86 of 109, or 79%—see Table 3, Panel A).[16] We restrict all further analyses to the 86 "non-contaminated" firms to focus our attention on the effects of auditor resignations without the confounding of other firm announcements.

3.2.2. Measurement of abnormal returns

Daily prediction errors in the event period are estimated for each sample firm as market adjusted returns. Specifically, we compute prediction errors for firm i in each period of T days as follows:

$$\text{Market Adjusted Pred. Errors}_{iT} = \begin{pmatrix} T \\ \Pi(1+R_{it})-1 \\ t=1 \end{pmatrix} - \begin{pmatrix} T \\ \Pi(1+R_{mt})-1 \\ t=1 \end{pmatrix} \tag{1}$$

where R_{it}, is the return for firm i on day t and R_{mt} is the equally weighted CRSP index on day t.[17]

The prediction errors are averaged across N firms in the sample for each period analyzed (hereafter, $\text{CAPE}_{t,t+k}$), where k is the number of days in the event window. To evaluate the statistical significance of CAPE in the event period, we report (1) t-statistics based on cross-sectional variance, (2) t-statistics based on the time-series variance of portfolio average prediction errors, and (3) p-values from sign tests. The t-statistic based on the time-series variance of portfolio average prediction errors, s^2_{PAPE}, estimated in the 200 days from day -500 to day -301 (or $+61$ to $+260$), is given by

[15] Using a similar procedure, we also evaluate the completeness of the *Auditor Trak* database. First, we conduct the above-specified search without explicitly including a company name. We identify 17 additional firms and establish that these firms are foreign firms (primarily Canadian and Australian firms that are not traded on US exchanges). We also perform a similar search on the 8-K sub-library of the COMPNY library in Lexis/Nexis. This procedure did not identify any additional firms. These tests increase the likelihood that our sample represents the population of firms with stock return data whose auditors resigned in the period from January 1994 to February 1998.

[16] The primary reason for contamination is the concurrent disclosure of earnings news, the secondary reason is the concurrent disclosure of management turnover news, and other reasons include the concurrent disclosure of news concerning debt covenants and lawsuits.

[17] We also estimated daily prediction errors for each sample firm based on (1) size-adjusted returns and (2) deviations of actual returns from market model predictions. Results for these alternative estimates are qualitatively the same as the results reported below for market adjusted returns.

Table 3
Frequencies of former client resignation characteristics

Panel A: Contamination[a]	
Total sample	109
Minus: Contaminated observations	23
Uncontaminated observations (i.e., sample of study)	86
Panel B: Frequencies of resignation characteristics for non-contaminated former clients	
Reason for resignation[b]	
Accounting concerns	21
No reason disclosed	65
Sample of study	86
Announcement source[c]	
Media	26
8-K	60
Sample of study	86
Prior audit opinion[d]	
Going concern	21
Other	65
Sample of study	86
Resignation timing[e]	
During audit period	35
Outside audit period	51
Sample of study	86
Auditor match[f]	
Mismatch	29
Appropriate match	33
Sample of study	62

[a] We consider a resignation announcement to be contaminated if a detailed DJNRS search for a given firm revealed potentially value relevant news items about that firm (e.g., earnings news, change in top management or board of directors, defaults, litigation against the firm) in the period −1 through +1.

[b] See Table 1.

[c] We used the earlier date of the 8K filing date and the media announcement date (if any) as our event date.

[d] We read the auditor reports in the 10K filings immediately preceding resignation to determine whether the prior audit opinion contained a going concern paragraph or not.

[e] We labeled resignations occurring during the last month of the fiscal year or after the fiscal year-end as resignations occurring during the audit period. We labeled resignations occurring between the last audit and the last month of the fiscal year as resignations occurring outside the audit period.

[f] We used the procedure developed by Shu (2000) to determine whether or not a former client was mismatched with the resigning auditor (see her paper for details on this procedure). Only 62 of 86 firms had sufficient data necessary to use the Shu procedure.

$$t = \mathrm{CAPE}_{t,t+k} \Big/ \sqrt{ks^2_{\mathrm{PAPE}}} \tag{2}$$

where PAPE_t is the portfolio average prediction error across N firms in the estimation period on day t.

3.2.3. Cross-sectional variation

We consider cross-sectional variation in stock price responses to resignation announcements by identifying several characteristics of the resignations. The frequencies of these characteristics for the "non-contaminated" observations used in the subsequent analyses are summarized in Table 3. Panel B of Table 3 reports frequencies of characteristics within the non-contaminated sub-sample. As described earlier, we identified the reason for the auditor's resignation by examining Forms 8-K and media reports related to each resignation. The proportion of resignations attributed to accounting concerns is 24% (21 firms) and is comparable to the proportion for the full set of resignation firms (i.e., 28% for contaminated and non-contaminated firms as reported in Table 1).

The earliest public disclosure of the auditor resignation is found in media reports for 26 (30%) of the non-contaminated observations. For 21 (24%) of the non-contaminated firms, the audit report immediately preceding the auditor resignation contains a going concern uncertainty paragraph in the independent auditor's report, and 35 (41%) of the firms experienced the audit resignation during the audit period, which we define as ranging from the last month of the fiscal year through the third month following the fiscal year end. Finally, we used the procedure developed by Shu (2000) to determine whether or not a former client was appropriately matched, in terms of Big 6 versus non-Big 6, with the resigning auditor. There are 62 firms in our sample for which sufficient data is available to perform the Shu (2000) analysis. We identified 29 firms (47%) as being inappropriately matched with their auditor.

4. Empirical findings

Our discussion of empirical results proceeds as follows. First, we evaluate the effect of auditor resignations on the stock prices of former clients. Our investigation includes analyses of the non-contaminated former clients and partitions based on the characteristics detailed in Table 3. The second section of empirical results provides evidence on the effect of auditor resignations on the stock prices of *continuing* clients (i.e., firms matched to former clients by industry, auditor and recent stock price performance) to assess whether the stock market interprets auditor resignations as positive news about the quality of continuing clients' financial disclosures. Finally, we present evidence on the stock-price reactions of firms matched by industry, and recent stock price

performance (but *not* auditor) to determine whether the price reaction for continuing clients reflects increased credibility of continuing clients' financial disclosures, or the positive effects of a potential decline in competition.

4.1. Former clients' stock market reaction to resignation announcements

Table 4 reports market-adjusted average prediction errors (CAPE) over various intervals, the earliest of which begins 300 trading days before the resignation announcement and the latest of which ends 60 trading days afterwards. Prior to the announcement, firms in our total sample experience negative abnormal performance; CAPE for the whole sample is −36.31%. This is consistent with prior research evidence suggesting that auditors resign from engagements when the financial condition of the client deteriorates (e.g., DeFond et al., 1997; Krishnan and Krishnan, 1997).

The average reaction of former clients' stock prices to initial public disclosure of auditor resignations is reliably negative, suggesting that disclosures of auditors' resignation actions convey incremental bad news about the former clients to investors. For the whole sample, the average abnormal return for days −1 through +1 is −5.57%. While the negative market reaction is consistent with prior auditor resignation research, it is premature to conclude that resignation announcements are bad news because simultaneous releases of other information create an event identification problem.

4.1.1. Non-contaminated resignation announcements

To increase the power of our hypotheses tests, we partition our sample based on whether the media reported contemporaneous (contaminating) news for the former client during the 3-day window surrounding the resignation. The results for the non-contaminated sub-sample in Table 4 is qualitatively similar to the whole sample results, but the stock returns in days −1 to +1, while reliably negative, are substantially less negative (i.e., −3.65% versus −5.57% for the whole sample). This underscores the importance of controlling for contaminating, concurrent news in the investigation of stock price reactions to auditor resignations.

Hypothesis H1a predicts that *attributed* auditor-resignation announcements (i.e., those announcements accompanied by information about reportable events) will be associated with negative abnormal returns for former clients at the time of the resignation announcement. Consistent with this prediction, and with findings in Whisenant et al. (2003), Table 4 reports that the average price reaction for the −1 to +1 day window is significantly negative (−9.71%, median = −5.81%) when there is explicit mention of an accounting disagreement or concerns over the reliability of the firms' internal controls.

In contrast, Hypothesis H1b predicts that *unexplained* auditor-resignation announcements (i.e., those not accompanied by information about reportable

Table 4
Univariate statistics of former clients' market-adjusted abnormal returns, in the pre- and post-event period windows, and in the event window

		Abnormal return window[a]								
		(−300, −61)	(−60, −2)	(2, 60)	(−1, 1)					
	N	Mean %AR[b]	Mean %AR	Mean %AR	Mean %AR	Median% AR	Cross-sectional t	Time-series t[c]	% of obs < 0	Sign test p-value
Benchmark samples										
Total sample	109	−36.31*	−16.66*	−11.37*	−5.57	−2.43	−3.59	−4.18	64.22	0.002
Sample of study	86	−32.99**	−13.41*	−11.35*	−3.65	−1.55	−2.69	−2.37	62.79	0.012
Reason for resignation										
Accounting concerns	21	−21.23	−24.64*	−5.69	−9.71	−5.81	−2.86	−2.72	71.43	0.039
No reason disclosed	65	−36.98***	−9.79**	−13.18**	−1.69	−1.03	−1.25	−1.08	60.00	0.068
Announcement source										
Media	26	19.32	−18.74*	−18.82*	−8.18	−5.08	−2.77	−3.02	65.38	0.084
8-K	60	−55.54*	−11.11**	−8.11	−1.69	−1.15	−1.20	−0.99	61.67	0.046
Prior audit opinion										
Going concern	21	−68.52*	−10.94	−16.16	1.70	2.37	0.92	0.51	38.10	0.134
Other	65	−20.96	−14.21*	−9.80**	−5.38	−3.39	−3.27	−3.14	70.77	0.001
Resignation timing										
During audit period	35	−42.11*	−16.64*	−11.26***	−4.63	−4.33	−2.07	−1.96	62.86	0.088
Outside audit period	51	−26.67	−11.20**	−11.41**	−2.98	−1.55	−1.75	−1.57	62.75	0.046

Auditor match										
Mismatch	29	0.60	−18.61*	−7.10	−2.23	−1.28	−1.05	−0.92	58.62	0.229
Appropriate match	33	−38.36*	−10.34**	−4.40	−6.82	−3.39	−2.95	−2.72	72.73	0.006

*,**,*** denotes significance at the 0.01, 0.05, and 0.10 two-tailed level of significance using a cross-sectional t-test, respectively.

[a] Trading days are relative to the date of the resignation announcement (day 0).

[b] Mean %AR is the cross-sectional average (across N firms) of market-adjusted abnormal returns (in %) for the period in question. Firm-specific abnormal returns for each window of T days are computed as:

$$\begin{pmatrix} T \\ \Pi(1+R_{it})-1 \\ t=1 \end{pmatrix} - \begin{pmatrix} T \\ \Pi(1+R_{mt})-1 \\ t=1 \end{pmatrix} \tag{1}$$

where R_{it} is firm i's raw return on day t, and R_{mt} is the CRSP equally weighted index on day t.

[c] Time-series t-statistics are computed as:

$$t = \mathrm{CAPE}_{t,t+k} \Big/ \sqrt{k s^2_{\mathrm{PAPE}}} \tag{2}$$

where $\mathrm{CAPE}_{t,t+k}$ is the mean prediction error in the event window, k is the number of days in the event window, and s^2_{PAPE} is the variance of portfolio average prediction errors over the estimation period (−500, −301 or +61, +260), where $PAPE_t$ is the portfolio average prediction error across N firms in the estimation period on day t.

events) will not be associated with abnormal returns for former clients at the time of the resignation announcement. Consistent with this prediction, Table 4 reports that the stock price reaction for uncontaminated unexplained resignation announcements is not statistically different from zero. Specifically, when no reason is disclosed, the average return is −1.69 with corresponding t-values of −1.25 for the cross-sectional t-statistic and −1.08 for the time series t-statistic. Further, the mean and median abnormal returns for attributed resignations are each significantly lower than those for unexplained resignations ($p = 0.02$ for the mean and median tests, not tabulated). Taken together, these results suggest that the presence or absence of a disclosure of concerns by the auditor, about the quality of former clients' financial reporting, is important to investors as they interpret the auditor resignation.

In additional partitions of the data reported in Table 4, we find that the average stock price reaction is significantly negative when the resignation is first disclosed in the news media (−8.18%, median = −5.08%). Resignations initially announced in 8-Ks do not lead to significantly negative abnormal returns in the −1 to +1 window and are significantly higher than the average returns for media disclosure firms ($p < 0.055$, two-tailed, not tabulated). Although the findings for 8-Ks could be due to slow dissemination of information contained in the 8-K report (see Dopuch et al., 1986), we also examine the stock price reaction during the period 0 through +4 days and find no significant abnormal returns for the 8-K sub-sample.

If the news-media imposes a filtering process to choose items deemed newsworthy, attributed resignations should be more frequently reported in the media. We find that out of 21 observations indicating explicit accounting concerns, 11 (52%) initially are disclosed in the news media. This proportion is significantly greater than the 15 out of 65 cases (23%) observed where the resignation is initially disclosed in the media, but no explicit reason is provided.

Results in Table 4 also indicate that firms that have *not* received a going concern uncertainty paragraph in the audit opinion preceding the resignation, experience a significantly negative event window return (−5.38%) The average abnormal return is significantly less than that for firms that *did* receive going concern explanatory language in the prior year's audit report ($p < 0.01$, two-tailed, not tabulated). It appears that an auditor's prior going concern opinion preempts the information content of resignation announcements.

We examine whether the stock price reaction differs depending on whether the resignation occurs during or after the audit period. Our expectation is that the abnormal returns would be more adverse for resignations occurring during the audit period, reflecting a greater likelihood that problems are identified during the engagement fieldwork, whereas resignations at other times of the year might be more reflective of the risk-screening process employed by audit firms. We find that firms with resignations during the audit period have, on average, a −4.63% abnormal return, which is reliably negative, but not

significantly different from the −2.98% abnormal return for firms with resignations outside the audit period. Investors apparently do not distinguish between resignations during the audit period and resignations at other times.

Finally, we examine whether the stock price reaction varies as a function of a proxy for auditor–client mismatch. Shu (2000) develops a model that predicts the likelihood that a client is associated with a Big 6 auditor.[18] We use this model to identify auditor–client mismatches in our sample. If market participants believe that an auditor and client are mismatched and anticipate the auditor's resignation, the abnormal return to the resignation announcement should be indistinguishable from zero. Alternatively, the stock price reaction to the resignation announcement should be more pronounced for clients that were identified as being appropriately matched with their auditor.

In Table 4, we report the market-adjusted abnormal returns in the event window for 62 firms out of 86 non-contaminated resignations for which sufficient available data exists to estimate Shu's (2000) model. The average stock-price reaction for the subset of firms that *are* appropriately matched with their auditors is reliably different from zero (mean = −6.82%), but it is not distinguishable from the average reaction for firms identified as mismatches (−2.23%) ($p = 0.159$, one-tailed, not tabulated).

4.1.2. Multivariate analysis of resignation characteristics

The previous section provides analyses of whether sample partitions differ in price responses for the resignation characteristics in Table 4. In Table 5, we present multivariate analyses of the price responses to resignations by including dummy variables for the characteristics noted in Table 4.

Regression (1) is estimated using all 86 non-contaminated observations of resignations from former clients. The regression estimate for the accounting concern dummy variable is significantly negative ($p < 0.025$, one-tailed), indicating that even when we include the other former client characteristics, the disclosure of an accounting concern at the time of the resignation disclosure has a significant effect on stock prices. The regression estimate for the effect of a going-concern uncertainty modification in the preceding audit opinion is marginally significantly positive ($p = 0.100$, two-tailed) suggesting that a prior going-concern uncertainty opinion mitigates the negative price response to a subsequent resignation.

[18] The dependent variable in Shu's (2000) model is a categorical variable indicating large auditors with 1 and zero otherwise. The five independent variables are firm size, acquisition expenditures, proceeds from raising outside capital, profitability, and market-to-book ratio. We estimate this model annually from 1993 through 1996, using all firms with available data on Compustat. We use the coefficient estimates to make one-year-ahead predictions in the period from 1994 through 1997. We identify, every year, the probability cutoff that minimizes the expected cost of misclassification. We use the model's predictions to determine whether firms are with the appropriate-sized auditor.

Table 5
Multivariate analyses of the former client's market-adjusted prediction errors during the event window[a]

		Intercept	Acctg-Concern[b]	Media[b]	Acctg-Concern * Media[b]	Going-Concern	Audit period	Mismatch	Adjusted-R^2
(1)	$n = 86$	−0.0246	−0.0631	−0.0361		0.0534	0.0034		9.23
		(0.290)	(0.025)	(0.115)		(0.100)	(0.922)		
(2)	$n = 86$	−0.0370	−0.0198	−0.0026	−0.1020	0.0598	0.0114		10.94
		(0.130)	(0.316)	(0.471)	(0.056)	(0.066)	(0.682)		
Test: Acctg-Concern + Media + Acctg-Concern * Media = 0, F-stat = 9.43, p-value = 0.002									
(3)	$n = 62$	−0.0430	−0.0499	−0.0265		0.0496	−0.0263	0.0421	7.67
		(0.147)	(0.082)	(0.221)		(0.212)	(0.440)	(0.196)	
(4)	$n = 62$	−0.0530	−0.0128	0.0061	−0.0905	0.0565	−0.0140	0.0321	8.50
		(0.085)	(0.392)	(0.445)	(0.113)	(0.158)	(0.698)	(0.334)	
Test: Acctg-Concern + Media + Acctg-Concern * Media = 0, F-stat = 4.37, p-value = 0.021									

[a] In this table we report estimation results from the following OLS regression model (p-values are reported in parentheses):

$$\text{CAPE}_{(-1,+1)} = \beta_0 + \beta_1\text{Acctg-Concern} + \beta_2\text{Media} + \beta_3\text{Acctg-Concern} * \text{Media} + \beta_4\text{Going-Concern} + \beta_5\text{Audit Period} + \beta_6\text{Mismatch} + \varepsilon.$$

"$\text{CAPE}_{(-1,+1)}$" equals the former client's market-adjusted prediction error over the window (−1, +1).
"Acctg-Concern" is a dummy variable that equals 1 when the reason for resignation was an accounting concern, and 0 if there was no reason for resignation disclosed.
"Media" is a dummy variable that equals 1 when the resignation was announced in the media, and 0 if it was first announced in the 8-K report.
"Going-Concern" is a dummy variable that equals 1 when the last audit opinion preceding resignation contained a going-concern paragraph, and 0 otherwise.
"Audit period" is a dummy variable that equals 1 when the resignation occurred during the audit period, and 0 otherwise. We labeled resignations occurring during the last month of the fiscal year or after the fiscal year-end as resignations occurring during the audit period.
"Mismatch" is a dummy variable that equals 1 when the former client and resigning auditor are classified as a mismatch using the Shu (2000) procedure, and 0 when they are not classified as a mismatch.

[b] Reported p-values reflect two-tailed tests except for the coefficients on variables Acctg-Concern, Media, and Acctg-Concern * Media.

In regression (2) we add an interaction dummy for the existence of an accounting concern and media disclosure to regression (1). In this specification, we find that neither of the two main effects is statistically significant, but that the coefficient estimate on the interaction variable is significantly negative at the 0.056 level. The combined estimated effect of a media disclosure and an accounting concern is significantly negative ($p = 0.002$). In this expanded regression the estimated coefficient for the going-concern uncertainty dummy variable is also significantly positive at the 0.066 level.

We assess the sensitivity of the results to influential observations in two ways. First, we eliminate the extreme 5% of observations on the dependent variable in each regression. Second, we eliminate all observations with studentized residuals greater than 3 in absolute value. The only qualitative change in results that we observe is that the coefficient estimate on the interaction term in regression (2) is no longer significant.

Regressions (3) and (4) are estimated with the 62 observations in our sample for which we have sufficient data to compute the Shu (2000) "mismatch" dummy variable. The estimated coefficient for the existence of an accounting concern dummy is marginally significant at the 0.082 level in regression (3), but not in regression (4) where we add an interaction dummy for the existence of an accounting concern and media disclosure. However, the combined effect of the accounting concern and media disclosure (i.e., the sum of the regression estimates for the two main effect and one interaction variable) in regression (2) is significantly negative ($p = 0.021$).

4.2. Continuing clients' stock market reaction to resignation announcements

We evaluate whether resignations provide incremental information about the quality of *continuing* clients by assessing whether resignation announcements positively affect the stock price of the auditors' continuing clients. We focus on the subset of the auditors' continuing clients for which we are most likely to observe an effect by selecting a sample of firms matched to former clients on industry and auditor as well as on prior stock price performance (hereafter continuing client matches). Thus, our analysis is based on firms that also experienced poor stock price performance prior to the resignation announcement. For such firms, a positive price reaction to the resignation announcement can be interpreted as an indication that investors view the resignation as a signal from the auditor, that risk factors that were associated with the client from which the auditor resigned, are apparently not as pervasive for the continuing client. Alternatively, to the extent that the resignation announcement signals poorer prospects for the former client, a positive price impact for the continuing client can result from reduced competition in the industry. We investigate this alternative explanation by collecting an additional sample of firms, matched to former clients on industry and prior stock price performance, but which are audited by a *different* auditor

(hereafter industry matches). Observing a positive price effect in both samples would make it more likely that the observed reaction is due to a reduction in competition, whereas observing a positive price reaction for the continuing clients, but not for the industry matches, would be consistent with our explanation that the resignation from the former client has led the stock market to improve its assessment of the quality of the continuing clients' financial disclosures.

Table 6 presents the frequencies of former client resignation characteristics for the 59 continuing clients matches (i.e., Table 6 shows the frequencies of the characteristics for the associated resignation firms). The relative frequencies of the characteristics in Panel B of Table 6 are similar to those of Table 3, suggesting that the identification of continuing clients does not systematically exclude any particular characteristic from the sample.

Table 7 presents market adjusted prediction errors (CAPE) for the event window $(-1, 1)$. The event window results suggest (1) a weak positive abnormal return of 1.52% for the whole sample of continuing clients (both contaminated and non-contaminated observations), (2) no abnormal returns for the non-contaminated sample firms (1.03% with a cross-sectional t-statistic of 1.01), (3) no abnormal returns for the (non-contaminated) continuing clients when the resignation is attributed to a reportable event, and (4) significantly positive abnormal returns for the (non-contaminated) continuing clients when the resignation is initially disclosed in the media (5.04%, significant at the 5% level for a one-tailed test).

Consistent with H2a, we find that *attributed* resignations provide no information about continuing clients: the mean 3-day abnormal return of −0.79% is not distinguishable from zero. On the other hand, we do not find support for H2b: the continuing clients' mean 3-day abnormal return of 1.82%, associated with *unexplained* resignations, is also indistinguishable from zero. However, evidence of information transfer is critically dependent on the news about the resignation being disseminated. Indeed, the 3-day abnormal return of 1.82% breaks down into 8.68% (t-statistic = 2.95) for unexplained resignations announced in the *media*, and −0.39% (t-statistic = −0.40) for unexplained resignations announced in the 8-K report (results not elsewhere tabulated).

To distinguish between an increase in the market's perception of quality and reduced competition, we evaluate the stock price reaction to resignation announcements for industry matches (i.e., matched on all previously listed criteria, *except auditor*). Partial results are reported in Table 7 and suggest that there are no significant stock price reactions for the whole sample or for sub-samples based on media announcements or reason for resignation. Results for other characteristics of the auditor resignation (i.e., prior audit opinion, resignation timing and auditor–client match) are not shown in Panel B, but none of the sub-sample abnormal returns are significantly different from zero.

Table 8 examines the continuing clients' abnormal returns in multivariate analyses. In regression (1), for continuing clients, we see the influence of the

Table 6
Frequencies of former client resignation characteristics, for continuing clients matches

Panel A: Data attrition	
Total sample (of former clients)	109
Minus: No matching continuing client found[a]	26
Total sample of continuing clients	83
Contamination[b]	
Total sample of continuing clients	83
Minus: Contaminated continuing clients	8
Uncontaminated continuing clients	75
Minus: Contaminated former clients	16
Uncontaminated continuing clients for which former clients are also uncontaminated (i.e., Sample of study of continuing clients)	59
Panel B: Frequencies of resignations for non-contaminated former clients associated with continuing client matches	
Reason for resignation[b]	
Accounting concerns	18
No reason disclosed	41
Sample of study of continuing clients	59
Announcement source[b]	
Media	18
8-K	41
Sample of study of continuing clients	59
Prior audit opinion[b]	
Going concern	14
Other	45
Sample of study of continuing clients	59
Resignation timing[b]	
During audit period	22
Outside audit period	37
Sample of study of continuing clients	59
Auditor match[b,c]	
Mismatch	23
Appropriate match	24
Sample of study of continuing clients	47

[a] We are unable to find continuing client matches for 26 of the 109 former clients because 17 former clients are audited by an auditor in the "Other" category on Compustat (which contains several auditors and so prohibits finding a match by the same auditor), and there are 9 former clients for which we find no firm in the same industry (i.e., the same 2-digit SIC code) audited by the same auditor and with all necessary stock price information.

[b] See Table 3. Note that these characteristics describe the former clients.

[c] Only 47 of 59 firms had sufficient data necessary to use the Shu procedure to determine whether a client was properly matched with its auditor.

Table 7
Univariate statistics of market-adjusted abnormal returns in event window (−1, 1) for continuing clients (Panel A), and for firms in the same industry as former clients but with a *different* auditor (Panel B)

	N	Mean %AR	Median % AR	Cross-sectional *t*	Time-series *t*[a]	% of obs < 0	Sign test *p*-value
Panel A: Continuing clients							
Benchmark samples:							
Total sample	83	1.52	0.09	1.74	1.52	45.78	0.255
Sample of study	59	1.03	0.03	1.01	0.82	47.46	0.397
Reason for resignation							
Accounting concerns	18	−0.79	−0.09	−0.68	−0.45	50.00	0.500
No reason disclosed	41	1.82	0.03	1.35	1.13	46.34	0.377
Announcement source							
Media	18	5.04	2.09	2.00	2.00	16.67	0.001
8-K	41	−0.74	−0.99	−0.88	−0.53	60.98	0.105
Prior audit opinion							
Going concern	14	1.42	−0.98	0.67	0.54	57.14	0.395
Other	45	0.90	0.07	0.44	0.66	44.44	0.276
Resignation timing							
During audit period	22	2.59	1.36	1.24	1.15	31.82	0.066
Outside audit period	37	0.09	−0.63	0.09	0.06	56.76	0.257
Auditor match							
Mismatch	23	2.29	0.18	0.99	0.87	47.83	0.500
Appropriate match	24	−0.51	−0.80	−0.52	−0.37	58.33	0.277
Panel B: Firms matched to former clients but with a different auditor							
Benchmark samples:							
Sample of study	55	1.41	−0.29	1.18	0.95	50.91	0.500
Reason for resignation							
Accounting concerns	16	0.71	0.22	0.46	0.28	43.75	0.4018
No reason disclosed	39	1.70	−0.32	1.08	0.93	53.85	0.3747
Announcement source							
Media	16	0.19	0.22	0.10	0.07	43.75	0.4018
8-K	39	1.91	−0.32	1.25	1.04	53.85	0.3747

[a] See Table 4.

media announcement source reported earlier—the estimated coefficient for media is significantly positive. Disclosure of an accounting concern for the former client is associated with a negative effect on the continuing clients. However, estimating the regression with a dummy variable for the combination of accounting concern and media disclosure (regression 2), we find that the estimated coefficient for accounting concern is no longer significant, but that the coefficients for media is 0.097 ($p = 0.001$), and for the interaction −0.077

Table 8
Multivariate analyses of the market-adjusted prediction errors during the event window for continuing clients (Panel A), and for firms in the same industry as former clients but with a *different* auditor (Panel B)[a]

		Intercept	Acctg-Concern[b]	Media[b]	Acctg-Concern ∗ Media[b]	Going-Concern	Audit period	Mismatch	Adjusted-R^2
Panel A: Continuing clients									
(1)	$n = 59$	−0.0201	−0.0394	0.0682		0.0314	0.0343		15.58
		(0.241)	(0.033)	(0.001)		(0.198)	(0.100)		
(2)	$n = 59$	−0.0296	−0.0114	0.0971	−0.0768	0.0345	0.0392		19.17
		(0.095)	(0.328)	(0.001)	(0.036)	(0.150)	(0.058)		
Test: Acctg-Concern + Media + Acctg-Concern ∗ Media = 0, F-stat = 0.09, p-value = 0.760									
(3)	$n = 47$	−0.0246	−0.0433	0.0775		0.0263	0.0176	0.0221	14.37
		(0.256)	(0.045)	(0.002)		(0.384)	(0.521)	(0.374)	
(4)	$n = 47$	−0.0332	−0.0119	0.1160	−0.0892	0.0298	0.0267	0.0142	18.60
		(0.129)	(0.346)	(0.001)	(0.043)	(0.312)	(0.329)	(0.566)	
Test: Acctg-Concern + Media + Acctg-Concern ∗ Media = 0, F-stat = 0.20, p-value = 0.657									
Panel B: Firms matched to former clients but with a different auditor									
(5)	$n = 55$	0.0239	0.0082	−0.0251		0.0135	−0.0191		−3.22
		(0.252)	(0.384)	(0.185)		(0.668)	(0.448)		
(6)	$n = 55$	0.0286	−0.0098	−0.0433	0.0488	0.0133	−0.0211		−3.66
		(0.187)	(0.388)	(0.117)	(0.193)	(0.668)	(0.403)		
Test: Acctg-Concern + Media + Acctg-Concern ∗ Media = 0, F-stat = 0.01, p-value = 0.908									
(7)	$n = 44$	0.0294	−0.0146	−0.0103		0.0086	0.0069	−0.0189	−10.24
		(0.262)	(0.318)	(0.375)		(0.822)	(0.830)	(0.544)	
(8)	$n = 44$	0.0321	−0.0269	−0.0263	0.0357	0.0080	0.0039	−0.0161	−12.17
		(0.232)	(0.243)	(0.273)	(0.280)	(0.824)	(0.912)	(0.632)	
Test: Acctg-Concern + Media + Acctg-Concern ∗ Media = 0, F-stat = 0.19, p-value = 0.668									

[a] In this table we report estimation results from the following OLS regression model (p-values are reported in parentheses):

$$\text{CAPE}_{(-1,+1)} = \beta_0 + \beta_1\text{Acctg-Concern} + \beta_2\text{Media} + \beta_3\text{Acctg-Concern} * \text{Media} + \beta_4\text{Going-Concern} + \beta_5\text{Audit Period} + \beta_6\text{Mismatch} + \varepsilon.$$

See Table 5.

[b] Reported p-values reflect two-tailed tests except for the coefficients on variables Acctg-Concern, Media, and Acctg-Concern ∗ Media.

($p = 0.036$) are significantly different from zero. These results suggest that the disclosure of a resignation from a former client that is reported in the media has a positive effect on continuing clients' stock prices, unless the disclosure also contains a statement by the resigning auditor about an accounting concern, in which case the effect on continuing clients is not distinguishable from zero. Similar results (including the interaction effect of accounting concern and media) are observed when the auditor mismatch variable is included in regressions (3) and (4).

We assess the sensitivity of the results to influential observations in two ways. First, we eliminate the extreme 5% of observations on the dependent variable in each regression. Second, we eliminate all observations with studentized residuals greater than 3 in absolute value. The qualitative changes in results that we observe are that (1) the coefficient estimate on audit period is no longer significant when we eliminate observations with extreme studentized residuals, (2) the coefficient estimate on going concern becomes marginally, positively significant, and (3) the coefficient estimate on the interaction term is no longer significant.

Comparable regression estimates for industry-matches are reported in Panel B. None of the reported coefficients are significantly different from zero, nor do the effects of including an interaction term (for accounting concern and media) reflect any significance.

Taken together, our findings suggest that unexplained auditor resignations in our sample enable market participants to assess continuing clients of the same auditor more positively when the resignation is announced in the media. The positive price reaction does not appear to result from reduced competition in the former client's industry because we find no significant stock price reactions for firms comparable to the former client but audited by a different audit firm.

5. Conclusion

We identify the conditions under which auditor resignations should reveal incremental information to investors and test our predictions for two types of companies: (1) those from which the auditor resigns (i.e., former clients) and (2) those that have similar (poor) prior stock-price performance as the former clients but that continue as clients of the resigning auditor (i.e., continuing clients). We jointly investigate former and continuing clients because, in recent years, auditors have adopted a portfolio approach to risk management that includes centralized risk-based screening of new and existing clients. In this risk-management environment, an auditor's resignation and retention decisions for troubled clients in the same industry are likely to be contemporaneous and may

reveal private information about the subset of an auditor's clients that are similar to former clients but from which an auditor does not resign.

We test our predictions on a sample of 109 auditor resignations from publicly traded companies during the period 1994–1998. Interestingly, only about 25% of our sample of auditor resignations is accompanied by disclosure of disagreements over accounting treatment, concerns about the adequacy of internal controls, or an unwillingness to rely on management's representations (i.e., factors that point toward diminished reliability of the former client's financial statements; collectively referred to as reportable events). Consistent with our predictions, we find that negative stock returns are generally restricted to this subset of "attributed" resignations and that unexplained resignations do not result in negative abnormal returns for former clients in the window around the resignation announcement. We conjecture that unexplained resignations are not incrementally informative because former clients experience declining performance prior to the resignation, and skeptical investors have already discounted former-client prices as compensation for greater uncertainty about financial reporting quality.

In contrast, we propose that *unexplained* resignations can provide incremental information to investors about continuing clients because these resignations are most likely associated with auditors' unobservable periodic risk-screening processes. In particular, in an environment in which poorly performing, high-risk clients are screened contemporaneously, the absence of resignation should convey incremental information about the credibility of the financial reporting of continuing clients. Our stock market analysis reveals positive abnormal returns for continuing clients, but only when the unexplained resignation is disclosed in the news media. The evidence is consistent with a scenario where the auditor risk-review process provides information about the credibility of continuing clients' financial statement disclosures so long as the information about the resignation is widely disseminated. We rule out competition-based intra-industry information transfer as an alternative explanation for the positive abnormal returns by finding no significant abnormal returns for a sample of firms matched on all of the previously listed criteria except auditor.

Our results should be interpreted in light of the caveat that auditor resignations occur among firms experiencing deteriorating financial and market performance. Therefore, constraints on data availability limit the number of firms that survive in our sample and can restrict our inferences to the set of firms that are similar to our surviving firms. Given this caveat, our findings suggest that auditor resignations triggered by accounting system disputes provide information about former clients and that the auditor risk-review process provides information about continuing clients. In both cases, the information conveyed by resignations relates to the credibility of the firm's financial reporting.

Our findings have potential implications for researchers as auditor resignations become more frequent post-Sarbanes-Oxley. First, we suggest that researchers consider our conjecture that investors require a premium for holding the stock of poorly performing firms to compensate them for greater uncertainty about the quality of such firms' financial reporting, as a lens through which to interpret their results. Second, we suggest researchers question the assumption that resignations are bad news in favor of studying what information resignations are likely to be conveying. Third, we suggest that researchers "decontaminate" their stock returns analyses to precisely isolate the effect of resignations.

Our findings also have a potential implication for auditors, because auditors' portfolio approach to risk management tends to target certain (undisclosed) industries. To the extent auditors' resignations convey information about continuing clients in the same industry, auditors may wish to avoid stringing out a series of resignations from multiple clients in the same industry. This is because market participants might bid up the stock price of continuing clients only to drop them again when the auditors resign later. Thus, simultaneous action for all troubled clients in the same industry could mitigate potential litigation risks.

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